Research Log – Monte Carlo Simulation for Financial Systems

1. What is a Monte Carlo Simulation?

A Monte Carlo simulation is a computational technique used to model systems that contain uncertainty or randomness.

Instead of solving a problem analytically, the system is simulated thousands or millions of times using random inputs, and the distribution of outcomes is analyzed.

Example:

If we want to estimate the expected value $E|f(x)|$ of some function. Instead of solving the integral analytically, we approximate it by averaging many random samples:

$$E|f(x)| \approx \frac{1}{N}\sum_{i=1}^{N} f(X_i)$$

Where $X_i$are randomly sampled values.

As the number of simulations N increases, the estimate converges to the true expected value.

2. Why Monte Carlo Simulation is Important?

Many real-world systems cannot be solved with closed-form mathematics.

Examples Include:

Problem	Why Analytical Solution Fails
Option pricing with complex payoffs	Integrals become intractable
Portfolio risk estimation	Many correlated assets
Derivative pricing under stochastic volatility	No closed form
Path-dependent options	Payoff depends on full price

Monte Carlo methods solve this by approximating the solution through simulation.

Instead of solving the system exactly, we simulate many possible futures.

In finance specifically, Monte Carlo simulation is commonly used for:

a. Option pricing
b. Risk estimation (Value at Risk)
c. Portfolio simulation
d. Interest rate modeling

3. Conceptual Example

Suppose a stock price follows a stochastic process.

One commonly used model is Geometric Brownian Motion (GBM):

$$dS = \mu S dt + \sigma S dW$$

Where:

Variable	Meaning
$\boldsymbol{S}$	Stock Price
$\boldsymbol{\mu}$	Expected Return
$\boldsymbol{\sigma}$	Volatility
$\boldsymbol{dW}$	Brownian Motion

Instead of solving this continuously, we simulate the price evolution in small time steps.

Example path w.r.t. time:

Time: 0 → 1 → 2 → 3 → …

Price: 100 → 102 → 98 → 101 → …

Each step contains random noise.

By simulation thousands of these price paths, we can estimate outcomes such as:

a. Expected future price
b. Probability of loss
c. Option payoff distributions

OBJECTIVE

The objective of this project is to build a Monte Carlo simulation framework for financial price processes and study how randomness propagates through time.

Specifically, the project will:

1. Simulate stochastic price paths using Geometric Brownian Motion
2. Generate thousands to millions of simulated market scenarios
3. Analyze the distribution of outcomes
4. Study convergence behavior as the number of simulation increases
5. Evaluate computation performance of large-scale simulations

This will help develop intuition about:

1. Stochastic processes
2. Uncertainty propagation
3. Computational finance methods

PROBLEM SETUP

The simulation will model the future behavior of a financial asset whose price evolves under uncertainty. In real markets, asset prices are influenced by numerous unpredictable factors including news, macroeconomic conditions, and trading activities. As a result, price movements contain a significant stochastic component.

To approximate this behavior, the asset price will be modeled as a stochastic process. Specifically, the simulation will assume that the price follows Geometric Brownian Motion (GBM).

Under this model, price changes are driven by two components:

a. A deterministic drift term, representing expected return
b. A random diffusion term, representing volatility

The continuous-time model can be written as:

$$dS = \mu S dt + \sigma S dW$$

The Monte Carlo simulation will generate many possible future price paths by repeatedly sampling the random component of this process.

For the initial experiments, the simulation will model a representative equity asset rather than a specific company. The goal at this stage is to study the behavior of stochastic processes rather than to forecast the price of a particular security.

The asset will therefore be initialized with realistic but simplified parameters for return and volatility. Once the simulation framework is established, the model can later be extended to simulate specific financial assets using historical market data.

Important:

When we simulate **millions of price paths**, we are **<u>not</u>** *predicting the future price.*

We are estimating the **distribution of possible futures.**

LOG ENTRY 1 – Simulated a Single Stochastic Price Path

Purpose

The purpose of the this first experiment is to simulate a single financial asset price path using Geometric Brownian Motion (GBM).

At this stage, the objective is to build and verify the basic Monte Carlo framework. A single simulated path allows us to observe how randomness enters the model and how a financial price process evolves through time under uncertainty.

The experiment serves as the foundation for later stages, where the model will be extended to generate many simulated paths, analyze distributions of outcomes, and evaluate the computation performance at larger scales.

Model

The asset price is assumed to follow GBM, given by:

$$dS = \mu S dt + \sigma S dW$$

Where:

Variable	Meaning
$\boldsymbol{S}$	Stock Price
$\boldsymbol{\mu}$	Expected Return
$\boldsymbol{\sigma}$	Volatility
$\boldsymbol{dW}$	Brownian Motion

Since a computer cannot simulate the process continuously, the model is discretized into time steps.

The discrete-time update equation is:

$$S_{t+1} = S_t * \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_t\right]$$

Where:

a. Δt is the time step
b. $Z_t \sim N(0,1)$ is a standard normal random variable

This allows the price path to be simulated step by step.

Parameters Used

For this first test, the simulation uses a representative equity asset with simplified parameters:

a. Initial price S_0 = 100 (This is the starting value of the asset at time t=0)
b. Drift μ = 0.08 (Drift represents the expected average rate of return of the asset over time. It is the long-term directional tendency of the price. In our simulation, this corresponds to an expected annual return of 8%)
c. Volatility σ = 0.20 (Volatility represents the degree of randomness or variability in price movements. Higher volatility means larger price swings, wider spread of possible outcomes, and greater uncertainty. 20% annual volatility is typical for many equity assets)
d. Time horizon T = 1 year (Time horizon defines how long into the future the simulation runs)
e. Number of time steps N = 252 (Since a computer cannot simulate continuous time, the process must be broken into discrete time steps. 252 corresponds to the approximate number of trading days in a year. So, each step represents $\Delta t = \frac{T}{N}$ which is roughly one trading day)

This first simulation is meant to verify three things:

1. The GBM discretization has been implemented correctly
2. Randomness produces path variability over time
3. The resulting path behaves like a plausible stochastic asset trajectory

At this stage, only one path is generated, so no distributional conclusions are drawn yet. The focus is purely on confirming the mechanics of the model.

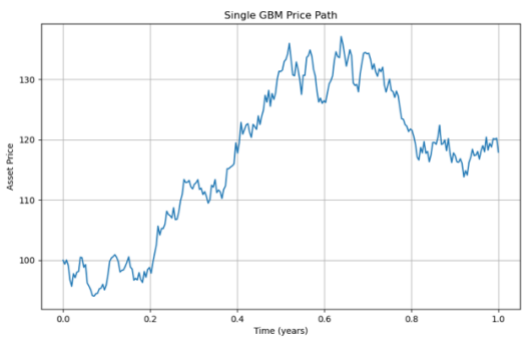
Observations

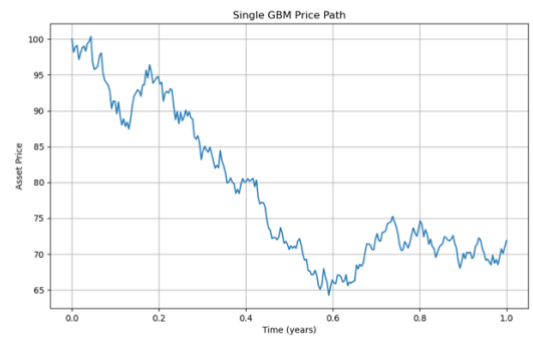
The simulation successfully generated a stochastic asset price path using the discretized Geometric Brownian Motion model. The asset price evolved continuously across the one-year horizon while exhibiting random fluctuations driven by normally distributed shocks at each time step.

Several properties consistent with the theoretical behavior of GBM were observed:

a. The asset price remained strictly positive throughout the simulation, which is expected due to the exponential formulation of the GBM update equation.
b. The price path exhibited random short-term fluctuations rather than a smooth deterministic trend.
c. Despite a positive drift parameter, the asset price experienced temporary drawdowns during the simulation.

Multiple runs of the simulation produced different price trajectories and final prices. This confirmed that the stochastic shocks generated at each time step introduce variability into the model, meaning each simulation represents a different possible realization of the underlying stochastic process.






```
Initial Price: 100.0
Final Price: 117.94914299762247
```

```
Initial Price: 100.0
Final Price: 71.88558335504688
```

Reflection

The first experiment validated that the discretized GBM implementation produces realistic stochastic price paths. The simulation behaves consistently with the theoretical properties of the model, including positivity of prices and variability across runs. With the basic framework verified, the next step is to extend the model to generate many simulated paths to analyze the distribution of outcomes rather than individual realizations.

LOG ENTRY 2 – Monte Carlo Simulation of Multiple Price Paths

Purpose

The purpose of this experiment is to extend the previous simulation from a single stochastic price path to many simulated paths to observe the distribution of possible outcomes.

While a single GBM trajectory represents one possible future evolution of the asset price, it does not provide information about the range of possible outcomes or their probabilities. Monte Carlo simulation addresses this limitation by generating a large number of independent realizations of the same stochastic process.

By simulating many price paths using the same model parameters, the experiment aims to approximate the probability distribution of the asset price at the end of the time horizon.

Goal

The experiment will generate multiple simulated price paths and analyze the resulting distribution of terminal prices.

Specifically, the experiment will:

1. Generate many independent GBM price paths using the same model parameters
2. Visualize multiple trajectories to observe how uncertainty evolves through time
3. Record the terminal price for each simulation
4. Analyze the distribution of terminal prices using summary statistics and visualizations

Model

In this experiment the same GBM model used in the previous simulation is applied repeatedly to generate many independent realizations of the asset price process. The only difference between simulations is the sequence of random shocks applied at each time step.

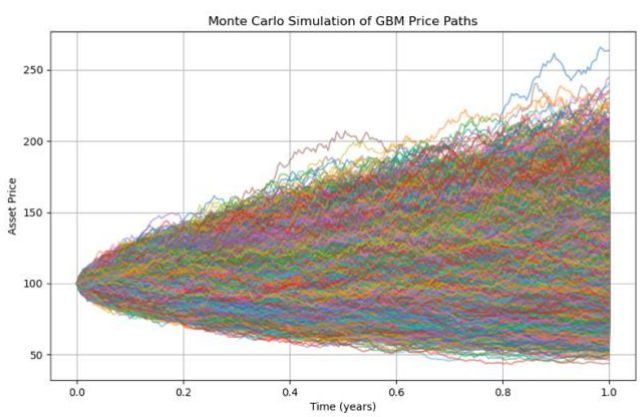
If the number of simulated paths is denoted by M, the simulation produces M independent realizations of the price process:

$$S_t^{(1)},\ S_t^{(2)},\ S_t^{(3)}, \dots,\ S_t^{(M)}$$

Each realization represents a different possible future evolution of the asset price under the same underlying model.

By generating many such paths, the Monte Carlo simulation approximates the probability distribution of the asset price at the terminal time T.

Observations




```
Number of simulations: 1000
Mean terminal price: 106.87025589419767
Min terminal price: 53.16897284913845
Max terminal price: 192.94800621232284
(.venv) (base) krishtanwar@Krishs-MacBook-Pro monte-
-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/
Number of simulations: 10000
Mean terminal price: 108.59412240429688
Min terminal price: 45.59540289346485
Max terminal price: 245.03403131870385
(.venv) (base) krishtanwar@Krishs-MacBook-Pro monte-
-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/
Number of simulations: 100000
Mean terminal price: 108.47872865167373
Min terminal price: 43.754908739783126
Max terminal price: 263.3755759571905
```

The Monte Carlo simulation generated many independent realizations of the GBM price process. All simulated paths began at the same initial price of 100, but the spread between trajectories increased over time. This widening dispersion reflects how uncertainty accumulates as random shocks compound throughout the simulation horizon.

The fan chart illustrates that while the expected drift of the process is positive, many individual price paths can experience both gains and losses. Many trajectories finished above the initial price, while others decline significantly, demonstrating that a positive drift parameter does not guarantee gains in any individual realization of the process.

As the number of simulated paths increased, the Monte Carlo estimate of the mean terminal price converged toward the theoretical expectation of the model. With 1000 simulations the estimated mean terminal price was approximately 1006.87. Increasing the simulation count to 10000 produced a mean of approximately 108.59, and with 100000 simulations the mean stabilized around 108.59. These values closely approach the theoretical expected value given by

$$E|S_T| = S_0 e^{\mu^T}$$

Which for the current parameters equals $100e^{0.08} \approx 108.33$. This behavior demonstrates the convergence properties predicted by the Law of Large Numbers in Monte Carlo simulations.

Increasing the number of simulations also revealed more extreme outcomes in the terminal price. For example, the minimum terminal price decreased from approximately 53 in the 1000-path simulation to about 43 in the 100000-path simulation, while the maximum terminal price increased from approximately 193 to about 263. This occurs because larger Monte Carlo samples are more likely to capture rare events in the tails of the distribution.

Finally, the spread of outcomes appears asymmetric, with the upside extending further than downside. This asymmetry is consistent with the log-normal distribution implied by the GBM model, where extreme positive outcomes are possible due to the multiplicative nature of price growth.

LOG ENTRY 3 – Terminal Price Distribution

Purpose

The purpose of this experiment is to analyze the statistical distribution of terminal asset prices generated by the Monte Carlo simulation.

While the previous experiment visualized the evolution of many simulated price paths, the focus now shifts to the distribution of outcomes at the terminal time. By examining the final price of the asset across all simulated paths, it is possible to observe the probability distribution implied by the GBM model.

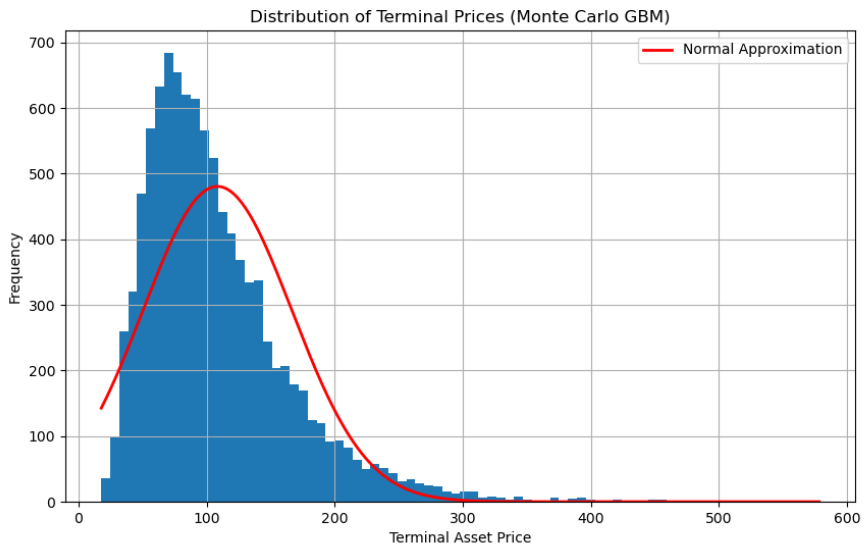
According to the GBM framework, the terminal prices of the asset should follow a log-normal distribution. This experiment therefore aims to visualize and verify this property using the results generated by the Monte Carlo simulation.

Goal

The experiment will analyze the terminal prices obtained from the Monte Carlo simulation by:

a. Plotting a histogram of terminal prices
b. Observing the shape of the distribution
c. Comparing the empirical mean with the theoretical expectation
d. Examining the presence of skewness and tail events

Observations



Note: In order to better visualize the skewness of the GBM terminal distribution, the volatility parameter was increased to $\sigma = 0.5$ *for this experiment.*

The histogram of terminal prices shows a strong concentration of outcomes around the central region near the mean, but the distribution extends much further on the right side than on the left. While the lower bound of the distribution is limited by the multiplicative nature of the price process, the upper tail stretches significantly further, allowing for rare but extremely large positive outcomes. This produces a clear right-skewed distribution.

The normal approximation overlaid on the histogram captures the central region of the distribution reasonably well but fails to match the heavier right tail. The simulated distribution therefore deviates from a true normal distribution, which is expected since the terminal price under GBM follows a log-normal distribution rather than a normal distribution.

LOG ENTRY 4 – 3D Visualization of Monte Carlo Price Paths

Purpose

The purpose of this experiment is to visualize the evolution of many simulated asset price paths in three dimensions in order to better understand how uncertainty propagates through time across multiple realizations of a stochastic process.

While previous experiments focused on individual paths and the distribution of terminal prices, this experiment aims to examine how the entire collection of simulated trajectories evolves simultaneously. By representing simulation index, time and price within a single visualization, it becomes possible to observe the structure of the stochastic surface produced by the Monte Carlo simulation.

Goal

The goal of this experiment is to simulate a large number of GBM price paths and visualize their behavior using a three-dimensional representation.

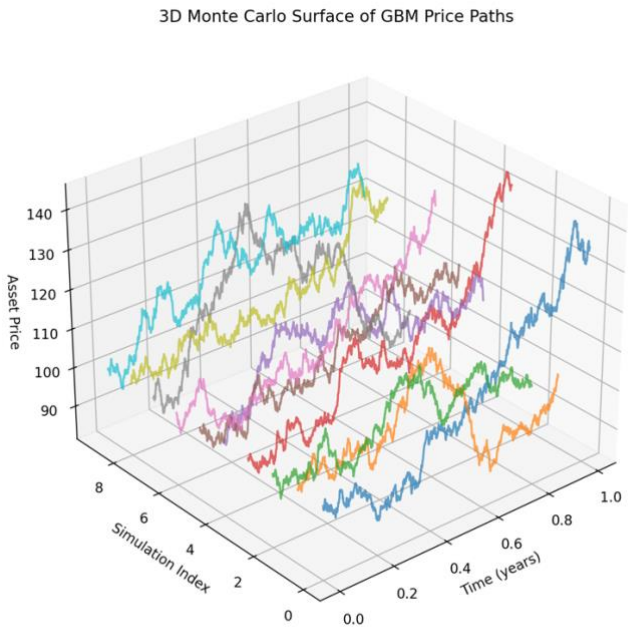
Specifically, the experiment aims to:

a. Generate a large Monte Carlo simulation of GBM price paths
b. Construct a three-dimensional visualization of the simulated trajectories
c. Observe how price dispersion grows as time progresses
d. Examine how multiple realizations of the stochastic process form a stochastic surface over time

Model

The simulation continues to use the Geometric Brownian Motion model for asset prices.

Observations



The three-dimensional visualization primarily serves as a conceptual tool for understanding how multiple stochastic trajectories evolve simultaneously. While the simulation index does not represent a financial variable, it allows the ensemble of simulated paths to be visualized as a surface evolving through time.

LOG ENTRY 5 – Probability Estimation from Monte Carlo Simulation

Purpose

The purpose of this experiment is to use the Monte Carlo simulation framework to estimate probabilistic properties of the asset price distribution.

In earlier experiments, the simulation was used to generate sample trajectories, observe path dispersion, and study the terminal distribution of the asset prices under the GBM model. However, the primary value of Monte Carlo methods in finance lies in their ability to estimate expectations, probabilities, and risk measures from a large number of simulated future scenarios.

Goal

The goal of this experiment is to extract useful probabilistic information from the Monte Carlo simulation of GBM price paths.

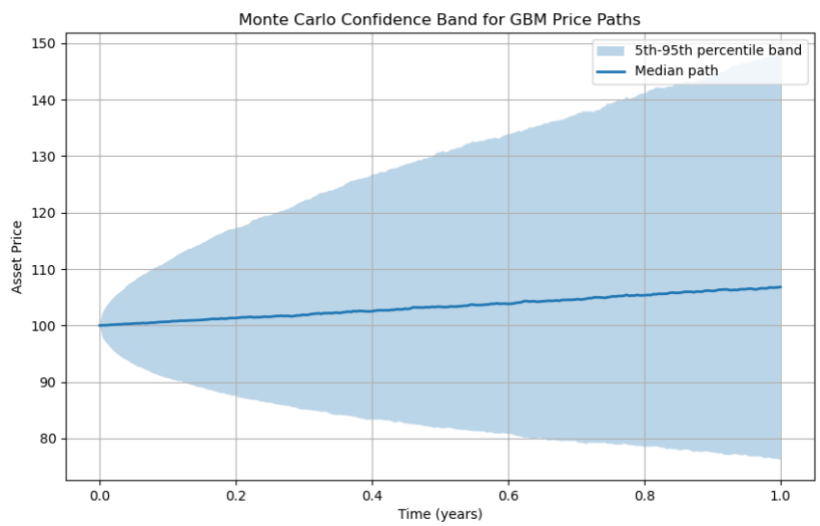
Specifically, the experiment aims to:

a. Estimate the expected terminal asset price
b. Estimate the probability that the terminal price exceeds the initial price
c. Estimate the probability of large positive or negative outcomes
d. Estimate confidence intervals for the terminal asset price
e. Visualize how the probability distribution evolves through time

Model

The simulation continues to use the GBM model for asset price.

Observations




```
Number of simulations: 10000
Expected terminal price (Monte Carlo): 108.75613930270339
Expected terminal price (Theory): 108.32870676749586
Probability terminal price > initial price: 0.6257
Probability terminal price > 120: 0.2805
Probability terminal price < 80: 0.0768
5th percentile terminal price: 76.1880449532964
Median terminal price: 106.80701284653787
95th percentile terminal price: 148.27609559707028
```

The Monte Carlo simulation produced a large sample of possible terminal asset prices after one year. From this simulated dataset, probabilistic quantities such as expectations, probabilities, and percentile ranges can be estimated.

Estimated Terminal Price

The expected terminal price estimated from the Monte Carlo simulation is given by:

$$E|S_T| = \frac{1}{N}\sum_{i=1}^{N} S_T^{(i)}$$

Using N = 1000 simulations, the estimated expectation was $E|S_T|$ = 108.76.

This value is close to the theoretical expectation of the GBM model:

$$E|S_T| = S_0 e^{\mu^T}$$

Where S_0 = 100, μ = 0.08, and T = 1 = 108.33

The small difference between the simulated estimate and the theoretical value is due to the sampling variability in the Monte Carlo simulation.

Probability Estimates

The simulation also allows estimation of probabilities of different outcomes.

For any event A, the Monte Carlo probability estimate is

$$P(A) = \frac{1}{N}\sum_{i=1}^{N} I_A(S_T^{(i)})$$

Where:

$$I_A = \begin{cases} 1 \; if \; the \; event \; occurs \\ 0 \; otherwise \end{cases}$$

Probability the price increases

The probability that the terminal price exceeds the initial price is estimated as:

$$P(S_T > S_0) \approx \frac{\#(S_T^{(i)} > S_0)}{N}$$

Using the simulated data this gives

$P(S_T > 100) \approx$ 0.6257

Indicating that approximately 62.6% of the simulated outcomes result in a price increase after one year.

Probability of a large gain

Similarly,

$$P(S_T > 120) \approx \frac{\#(S_T^{(i)} > 120)}{N}$$

Which gives

$P(S_T > 120) \approx$ 0.2805

This suggests that roughly 28% of the simulated scenarios produce a price greater than 120.

Probability of a large loss

$$P(S_T < 80) \approx \frac{\#(S_T^{(i)} < 80)}{N}$$

Which gives

$P(S_T < 80) \approx$ 0.0768

Only in 7.7% of the simulated cases the terminal price falls below the initial price.

Percentile Interpretation

Percentiles provide a probabilistic range for future prices.

If the simulated terminal prices are sorted

$$S_{(1)} < S_{(2)} < \cdots < S_{(N)}$$

The P-th percentile corresponds to the value

$S_{(k)}$ where $k = pN$

For $N = 1000$

5th percentile

$k = 0.05 * 10000 = 500$

The 5th percentile was

$S_{(500)} \approx$ 76.19

Which implies

$P(S_T < 76.19) \approx 0.05$

Meaning 5% of the simulated outcomes fall below this value.

Median (50th Percentile)

$k = 0.5 * 10000 = 5000$

The median was

$S_{(5000)} \approx$ 106.81

Indicating that half of all simulated prices fall below this value and half above it.

95th Percentile

$k = 0.95 * 10000 = 9500$

The value obtained was

$S_{(9500)} \approx 148.28$

Meaning that 95% of simulated prices fall below this value.

Confidence Interval Interpretation

Combining the 5th and 95th percentiles produce a probabilistic range:

$$76.19 < S_T < 148.28$$

This implies

$$P(76.19 < S_T < 148.28) \approx 0.90$$

Therefore, approximately 90% of simulated outcomes fall within this interval.

This interval represents the likely range of terminal prices implied by the GBM model under the assumed parameters.

Confidence Band Through Time

The confidence band plot shows the 5th and 95th percentiles of the simulated prices at each time step.

For each time t, the percentiles are computed as

$$S_{5\%}(t) = percentile_5(S_t^{(1)}, S_t^{(2)}, \dots, S_t^{(N)})$$

$$S_{95\%}(t) = percentile_{95}(S_t^{(1)}, S_t^{(2)}, \dots, S_t^{(N)})$$

The shaded region between these two curves represents the range containing approximately 90% of simulated price paths at each time step.

The widening of this band over time illustrates how uncertainty increases as stochastic shocks accumulate in the price process.

Interpretation

From a financial decision-making perspective, the Monte Carlo simulation provides a probabilistic view of the potential outcomes for the asset price over a one-year horizon. Assuming the current price is 100, the simulation estimates an expected terminal price of approximately 108.76, implying an expected return of roughly **8.76%.** This is consistent with the theoretical expectation of **108.33** predicted by the GBM model.

However, the simulation also highlights the uncertainty surrounding this expected outcome. The probability that the asset price finished above the current price is approximately **62.6%,** suggesting that the asset is more likely that not to appreciate over the one-year horizon. At the same time, the model indicates that there is still meaningful probability of adverse outcomes.

For example, the probability that the price exceeds 120 is approximately **28%,** representing a scenario in which the asset generates a return of at least **20%**. Conversely, there is approximately a **7.7%** probability that the price falls below 80, representing a potential loss of more than **20%.**

For an investment decision, these results highlight the trade-off between expected return and downside risk. The model suggests that the investment has a positive expected return and a greater than 60% probability of appreciation, but it also carries a non-negligible risk of significant losses. As a result, the decision to invest would depend on whether the expected upside and probability of gains are sufficient to compensate for the potential downside implied by the lower tail of the distribution.

LOG ENTRY 6 – Calibrating the Monte Carlo Model to Real Equities

Purpose

The purpose of this experiment is to move from a purely theoretical Monte Carlo simulation to a market-calibrated simulation based on real equity data.

In the earlier logs, the GBM model was explored using assumed values of drift and volatility in order to understand stochastic paths, Monte Carlo convergence, terminal distributions, and probabilistic risk ranges. While those experiments were useful for building intuition, the parameters used were hypothetical.

This experiment extends the framework by calibrating the model to two real equities I hold: Nvidia (NVDA) and Vodafone Idea (IDEA.NS). By estimating drift and volatility from historical market prices, the simulation becomes more representative of actual assets and allows the Monte Carlo framework to be interpreted in practical investment context.

Goal

The goal of this experiment is to estimate GBM parameters from historical price data for NVDA and IDEA.NS and then use those calibrated parameters to generate Monte Carlo simulations for each stock.

Specifically, this experiment aims to:

a. Download historical price data for NVDA and IDEA.NS
b. Compute historical log returns for each equity
c. Estimate drift μ and volatility σ from those returns
d. Simulate future price paths using the calibrated GBM model
e. Compare the probabilistic outlook for an American and an Indian equity using the same Monte Carlo framework

This makes the model more realistic because the simulated price dynamics are now linked to observed market behavior rather than assumed parameter values.

Model

The simulation continues to use the GBM model.

For this experiment, μ and σ are no longer assumed. Instead, they are estimated from historical daily log returns:

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$$

If the average daily log return is denoted by r and the standard deviation of daily log return is denoted by S_r, then the annualized estimates are:

$$\mu \approx 252r$$

$$\sigma \approx \sqrt{252}S_r$$

Assuming approximately 252 trading days per year.

These estimated values are then used as inputs to the Monte Carlo simulation for each stock.

Procedure

The procedure for this experiment is designed to calibrate the GBM model using historical data and then evaluate its predictive distribution against subsequently realized stock prices.

1. Collect historical data

Historical daily price data is obtained for each stock. The dataset is divided into two segments:

   a. Calibration period: all data up to December 31, 2024
   b. Validation period: data from January 1, 2025 onwards

The calibration period is used to estimate model parameters, while the validation period is reserved for comparing the simulation against actual realized prices.

2. Compute daily log returns

For each stock, daily log returns are calculated from the historical price series in the calibration period. These returns provide the empirical basis for estimating the drift and volatility of the GBM model.

3. Estimated GBM parameters

From the daily log return series, the following quantities are computed:

   a. The average daily log return
   b. The standard deviation of daily log returns

These are then annualized to obtain model parameters. Assuming 252 trading days a year.

4. Set initial simulation price

The last observed stock price in the calibration period, i.e. the closing price on December 31, 2025, is used as the initial price for the Monte Carlo simulation.

This ensures that the forecast begins from the same point at which the real validation period begins.

5. Generate Monte Carlo Paths

Using the calibrated values and the initial paths, a large number of future price oaths are simulated over the validation horizon using the GBM model.

The simulation generates an empirical distribution of possible future prices rather than a single predicted path.

6. Compute forecast bands

At each simulated time step, summary statistics of the Monte Carlo distribution are computed, including:

a. The median simulated price
b. The 5th percentile
c. The 95th percentile

These quantities form a probabilistic forecast band showing the range within which most simulated prices are expected to lie.

7. Overlay realized stock prices

The actual observed stock prices during the validation period are then plotted on the same graph as the Monte Carlo forecast band.

This allows the model to be evaluated visually by checking whether the realized price path remains within the simulated range and how closely it follows the central tendency of the forecast distribution.

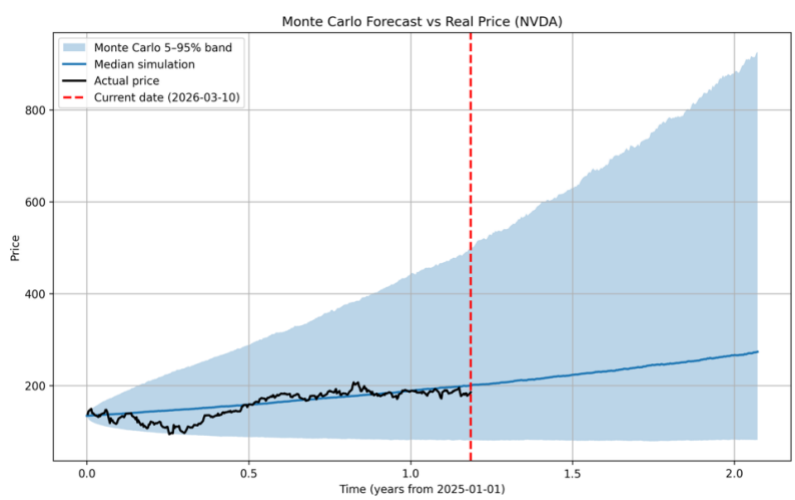
8. Compare the two equities

Finally, the calibrated parameters and forecast behavior of NVDA and IDEA.NS are compared in terms of:

a. Estimated drift
b. Estimated volatility
c. Width of simulation band
d. Relationship between the realized path and the forecast distribution

This comparison helps assess how the same GBM framework behaves when applied to two different real equities.

Observations




```
Estimated drift (mu): 0.47317103345524353
Estimated volatility (sigma): 0.514543010751158
Initial simulation price: 134.25328063964844
Expected terminal price (Monte Carlo): 234.98776822397951
Expected terminal price (Theory): 234.0456935583582
5th percentile terminal price: 80.57430855587575
Median terminal price: 200.6626410449232
95th percentile terminal price: 506.9589137923385
Probability terminal price > initial price: 0.7673
Probability terminal price > 1.2*S0: 0.6522
Probability terminal price < 0.8*S0: 0.1341
```

The objective of this experiment was to simulate future price paths for NVDA and IDEA.NS using a GBM model and evaluate how the simulated distribution compares with the realized price trajectory.

The model parameters (drift and volatility) were estimated using historical daily log returns from 2018 to 2024, and the simulation was then run forward starting from December 31, 2024.

A Monte Carlo simulation with 10000 price paths was used to estimate the distribution future outcomes.

From the results for **NVDA** we can see that the Monte Carlo mean closely matches the theoretical GBM expectation, confirming that the simulation numerically consistent with the analytical model.

From the simulated terminal price distribution:

$$P(S_T > S_0) = 83.26\%$$

$$P(S_T > 1.2S_0) = 76.4\%$$

$$P(S_T < 0.8S_0) = 10.6\%$$

These probabilities reflect the positive drift estimated from historical returns.

However, they should not be interpreted as predictions. These are model-implied probabilities under the GBM assumptions.

a. Realized price remained within the Monte Carlo confidence band

The actual NVDA price path stayed within the simulated 5-95% distribution band, indicating that the GBM model produces a distribution of outcomes consistent with the realized trajectory.

b. Median simulations tracked the realized price reasonably well

The realized price path remained closed to the median simulated trajectory for much of the validation period. This suggests that the estimated drift parameters produced a central forecast path broadly aligned with observed market behavior

c. Prediction interval expands rapidly

The uncertainty cone widens significantly over time, with the upper bound reaching above 900.

This is a direct consequence of the GBM assumptions that variance grows proportionally with time. As a result, long-horizon forecasts become extremely uncertain.

d. Distribution is strongly right skewed

The large difference between:

- Mean terminal price: 358
- Median terminal price: 276

Demonstrates the right-skewed nature of the lognormal distribution. A small number of extreme upward price paths increase the mean significantly relative to the median.

Model Limitations

This experiment highlights several limitations of the GBM framework when applied to individual equities:

1. Constant volatility assumption

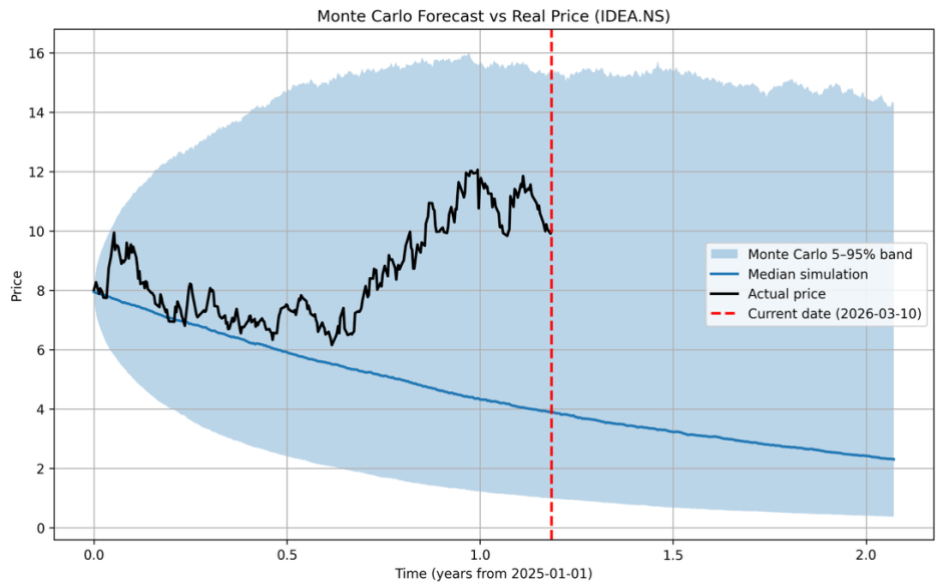
Real markets exhibit volatility clustering, whereas GBM assumes volatility remains constant over time.

2. Drift instability

The drift estimate is sensitive to the training period. The strong NVDA bull run during the training window estimated drift.

3. Long-horizon uncertainty

The widening prediction interval demonstrates that GBM becomes less informative for long-term forecast, as the range of plausible outcomes expands dramatically.




```
Estimated drift (mu): -0.30256656300816226
Estimated volatility (sigma): 0.7631047117461228
Initial simulation price: 7.940000057220459
Expected terminal price (Monte Carlo): 4.2460838048301115
Expected terminal price (Theory): 4.242565933057855
5th percentile terminal price: 0.37293308405707637
Median terminal price: 2.301251845246345
95th percentile terminal price: 14.304568393427354
Probability terminal price > initial price: 0.1272
Probability terminal price > 1.2*S0: 0.0972
Probability terminal price < 0.8*S0: 0.8287
```

For IDEA.NS, the realized price path from 2025 to early 2026 does not closely follow the median simulated trajectory.

Key Observations:

1. The median simulation trends downward

The median path declines steadily from about 7.94 toward much lower levels, reflecting the strongly negative estimated drift. This means the model's central expectation is bearish.

2. The actual price path is materially stronger than the median forecast.

The realized price path remains well above the simulated median for most of the validation period and even risks around 10-12 before the current date marker. This indicated that the actual market performance was much stronger than the central GBM forecast implied by the historical training sample.

3. The realized path still lies within the simulation band

Although the actual price path outperformed the median, it remains inside the broad 5-95% Monte Carlo band. This means the realized outcome is not inconsistent with the model distribution, but it lies in the stronger part of that range rather than near the center.

4. The prediction band is very wide

The upper end of the band remains high while the lower end approaches values close to zero. This reflects the combination of:

a. High volatility
b. A lognormal price process
c. Long forecast horizon

The result is a very broad range of possible outcomes, which reduces the precision of the forecast.

This experiment shows that the GBM model is internally consistent, but the fitted parameters produce a central forecast that does not align well with the realized trajectory of IDEA.NS over the validation window.

The negative drift estimate appears to be heavily influenced by the historical training period. In other words, the model extrapolates past weakness into the future, while the actual stock behavior over 2025-2026 turned out to be stronger than that historical trend would suggest.

This highlights that drift estimation is highly sensitive to the selected training window, especially for volatile equities.

During the analysis of IDEA.NS, the brokerage platform issued a regulatory caution highlighting two key financial concerns:

1. The company currently reports zero or negative earnings per share (EPS).
2. The company has been loss-making for the last eight consecutive quarters.

These warnings indicated that the firm has not generated sustained profitability in recent years and may face ongoing financial pressure. Such conditions often correspond to elevated uncertainty in the equity's valuation and can lead to higher volatility in market prices.

This risk profile is reflected in the statistical characteristics estimated from historical price data. The model estimated:

$$Drift\ (\mu) = -0.3026$$

$$Volatility\ (\sigma) = 0.7631$$

The negative drift suggests that the historical return series implies a downward long-term tendency in price under GBM framework. At the same time, the very high volatility produces a wide distribution of simulated outcomes.

Monte Carlo simulations produced the following terminal statistics:

a. Median terminal price: 2.30
b. 5th percentile: 0.37
c. 95th percentile: 14.30

The probability analysis also showed:

$$P(S_T > S_0) = 12.7\%$$

$$P(S_T < 0.8S_0) = 82.9\%$$

Under the model assumptions, the majority of simulated outcomes lie below the initial price, reflecting the historically negative drift.

However, the realized market price during the validation period remained significantly stronger than the simulated median path. This shows an important limitation of GBM-based forecasting – parameter estimates derived from historical data may not fully capture future market sentiment, structural changes, or speculative dynamics.

Conclusion

This project implemented a Monte Carlo simulation framework using Geometric Brownian Motion (GBM) to model potential future stock price paths based on historically estimated drift and volatility. The simulations produced price distributions whose expected values closely matched the theoretical GBM expectation, confirming the correctness of the implementation.

Comparisons between simulated paths and realized prices showed that while actual market movements generally remained within the simulated probability bands, they did not always follow the median forecast. This highlights both the usefulness of Monte Carlo simulations for understanding uncertainty and the limitations of constant-drift, constant-volatility models when applied to real-world financial markets.